

Total No. of Questions : 8]

[Total No. of Printed Pages : 2

Roll No

EX-6005(1)-CBGS

B.E. VI Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Utilization of Electrical Energy

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) State inverse square law and lamberts cosine law of illumination? 7
b) What is solid angle? What is flood lighting and where it is used? 7
2. a) What is dielectric heating? Explain factors on which the dielectric loss in a dielectric material depends? 7
b) What is Induction heating? Explain high frequency induction furnace with its advantages. 7
3. a) Explain dead weight, accelerating weight and train resistance referred to traction? 7
b) Explain why a DC series motor is ideally suited for traction purpose. 7
4. a) What is an electric drive? Why electrical drive produces noise? 7
b) Explain regenerative braking in electric motors? 7

EX-6005(1)-CBGS

PTO

[2]

5. a) Write short notes on Hybrid vehicle. 7
b) What type of transmission system used in electric vehicle? 7
6. a) What are the advantages of electric heating? 7
b) What are the different types of electric welding? 7
7. a) Explain what is mean by Load Equalization and how it is accomplished? 7
b) What do you mean by braking? Explain with example. 7
8. a) Explain terms: 7
i) Glare
ii) Reflection
iii) Solid angle
b) Explain in detail with the help of block diagram the operation of an electrical vehicle? 7

EX-6005(1)-CBGS